

Digital Circuits Homework 1

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Instructions

- **Write your solutions by hand (do not type them).** Show the necessary steps and clearly label all circuit diagrams. If the space below a problem is insufficient, you may attach additional sheets. Do not use AI tools to complete this homework.
- Unless otherwise specified, perform binary arithmetic directly in binary.
- In logic expressions, $+$ denotes OR, juxtaposition denotes AND, $'$ denotes NOT, and $\oplus$ denotes XOR.
- Upload your hand-written solutions to Blackboard as a **single PDF file**.
- Late submission may result in a score deduction.
- Total score: **100 points**.

Problem 1. (12 pts) **Check the conversions.**

A classmate wrote down the following conversions. For *each* statement, decide whether it is correct. If it is incorrect, replace the value on the right-hand side with the correct value. Show enough work to justify your decision. Each part is worth 4 points.

(1) $(101101)_2 = (44)_{10}$

11) $(101101)_2 = 2^0 + 2^2 + 2^3 + 2^5 = 1 + 4 + 8 + 32 = 45 \neq (44)_{10}$

$\therefore (101101)_2 = (45)_{10}$

(2) $(11010110)_2 = (327)_8$

12) $(110)_2 = 0 + 2^1 + 2^2 = 6$ $\therefore (11010110)_2 = (326)_8 \neq (327)_8$

$(010)_2 = 2^1 = 2$

$(011)_2 = 2^0 + 2^1 = 3$

(3) $(101.011)_2 = (5.375)_{10}$

13) $(101.011)_2 = 2^0 + 2^2 + 2^{-2} + 2^{-3} = \frac{1}{8} + \frac{1}{4} + 5 = \frac{3}{8} + 5 = (5.375)_{10}$

$\therefore (101.011)_2 = (5.375)_{10}$

Problem 2. (14 pts) Complete the binary calculations.

- (1) (7 pts) One bit of the first operand is missing:

$$(101?01)_2 + (011011)_2 = (1001000)_2.$$

Find the missing bit. Show the binary addition, including carries, and explain how you determined the bit.

$$\begin{array}{r} 101?01 \\ - 011011 \\ \hline 010101 \end{array}$$

$$\therefore ? = 1$$

The missing bit is 1.

$$(1001000)_2 = (101101)_2 + (011011)_2$$

- (2) (7 pts) Multiply directly in binary:

$$(1011)_2 \times (101)_2.$$

Show all partial products and give the result in binary.

$$(1011)_2 \times 1 = 1011$$

$$(1011)_2 \times 0 = 0000$$

$$(1011)_2 \times 1 = 1011$$

$$\begin{array}{r} \text{So: } 1011 \\ \times \quad 101 \\ \hline 1011 \\ 0000 \\ 1011 \\ \hline 11011 \end{array}$$

$$\therefore (1011)_2 \times (101)_2 = (11011)_2$$

$$2^0 \rightarrow 2^1 : \text{carry} = 1$$

$$2^1 \rightarrow 2^2 : \text{carry} = 1$$

$$2^2 \rightarrow 2^3 : \text{carry} = 1$$

$$2^3 \rightarrow 2^4 : \text{carry} = 1$$

$$2^4 \rightarrow 2^5 : \text{carry} = 1$$

$$2^5 \rightarrow 2^6 : \text{carry} = 1$$

Problem 3. (12 pts) **A signed-arithmetic check.**

A device uses **8-bit two's-complement** numbers. All numbers in the following expressions are decimal.

For each calculation, write both operands in 8-bit binary, perform the binary addition, give the final 8-bit result, and state whether **signed overflow** occurs. Each part is worth 6 points.

(1) $85 + 52$

$$(85)_{10} = 01010101 \quad \therefore 01010101 + 00110100 = 10001001$$

$$(52)_{10} = 00110100$$

$$\begin{array}{r} 01010101 \\ + 00110100 \\ \hline 10001001 \end{array}$$

$\therefore$ signed overflow occurs.

(2) $-70 + 25$

$$70 = 01000110$$

$$-70 = 10111010$$

$$25 = 00011001$$

$$\therefore -70 + 25 = 10111010 + 00011001$$

$$\begin{array}{r} 10111010 \\ + 00011001 \\ \hline 11010011 \end{array}$$

$$\therefore -70 + 25 = 11010011$$

signed overflow doesn't occur.

Problem 4. (12 pts) **Identify an unknown gate.**

A two-input device contains exactly one of these gates: **AND**, **OR**, **XOR**, or **NAND**. A technician recorded three measurements; one output value is missing.

A	B	Y
0	0	0
0	1	1
1	0	?
1	1	0

- (1) (3 pts) Identify the gate and fill in the missing output. Explain why the other three candidate gates do not fit the recorded measurements.

It's a XOR gate, $? = 1$

If it's an AND gate: $A=1, B=1, Y=1$

If it's an OR gate: $A=1, B=1, Y=1$

If it's a NAND gate: $A=0, B=0, Y=1$

- (2) (3 pts) Write a logic expression for Y using **only AND, OR, and NOT**. You may obtain the expression directly from the truth table; no simplification is required.

$$Y = \bar{A}B + A\bar{B}$$

- (3) (6 pts) Apply this gate independently to corresponding bits of two 8-bit inputs:

$$A = 10110110, \quad B = 01101101.$$

Show the bitwise calculation and write the 8-bit output.

2^0 : $A=0$	$B=1$	$Y=1$	$\therefore$ output is: 11011011
2^1 : $A=1$	$B=0$	$Y=1$	
2^2 : $A=1$	$B=1$	$Y=0$	
2^3 : $A=0$	$B=1$	$Y=1$	
2^4 : $A=1$	$B=0$	$Y=1$	
2^5 : $A=1$	$B=1$	$Y=0$	
2^6 : $A=0$	$B=1$	$Y=1$	
2^7 : $A=1$	$B=0$	$Y=1$	

Problem 5. (12 pts) **Compare three circuits by their outputs.**

Three circuits have the following input-output expressions:

$$F = (A + B)C, \quad G = AC + BC, \quad H = A + B + C.$$

Compare the circuits by evaluating their outputs for each input combination.

(1) (8 pts) Complete the truth table.

A	B	C	F	G	H
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	0	1
0	1	1	1	1	1
1	0	0	0	0	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	1	1	1	1

(2) (4 pts) Based on the completed table, which of the three circuits always produce the same output? Give one input combination on which another circuit produces a different output, and state the two output values.

circuit F and circuit G always produce the same output.

when $A=0, B=0, C=1$:

$$F = G = 0, \quad H = 1$$

Problem 6. (14 pts) **Fill in a missing circuit block.**

The block G below has two inputs, A and B , and one output. The final gate is a 2-input AND gate. The required output Y is specified by the table.



	A	B	C	Required Y
	0	0	0	0
✓	0	0	1	0
	0	1	0	0
✓	0	1	1	1
	1	0	0	0
✓	1	0	1	1
	1	1	0	0
✓	1	1	1	0

- (1) (4 pts) Complete the truth table of the missing block G . Explain which rows of the given table determine its output.

A	B	G -output
0	0	0
0	1	1
1	0	1
1	1	0

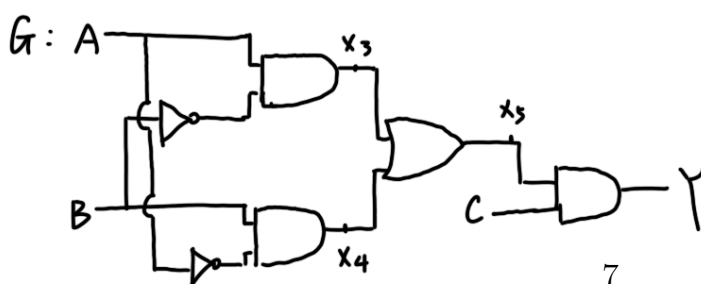
rows 2, 4, 6, 8 determine G 's output.

- (2) (4 pts) Name the gate that can serve as G and write its logic expression in terms of A and B .

G is a XOR gate

$$G = A\bar{B} + \bar{A}B$$

- (3) (6 pts) Suppose that gate is unavailable. Draw an implementation of G using **only** AND, OR, and NOT gates. Connect it to the final AND gate to obtain Y . Label the intermediate signals. No Boolean-algebra simplification is required.



$$\begin{aligned} x_1 &= \bar{B} \\ x_2 &= \bar{A} \\ x_3 &= A\bar{B} \\ x_4 &= \bar{A}B \\ x_5 &= A\bar{B} + \bar{A}B = G \end{aligned}$$

Problem 7. (14 pts) **Only NOR gates are available.**

A box contains only **2-input NOR gates**. Implement

$$\underline{Y = (A + B)C}$$

using **no more than three** 2-input NOR gates.

- (1) (5 pts) Specify the inputs and output of each NOR gate. Name every intermediate signal. Explain why the resulting output has the required value; you may reason directly from gate behavior.

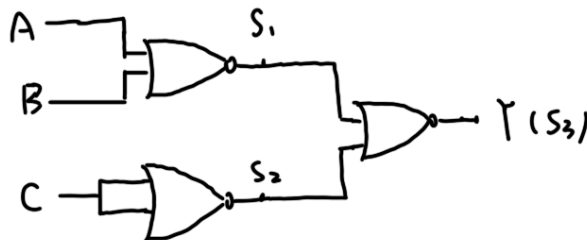
The first NOR gate has two input A and B, produces $S_1 = \overline{A+B}$

The second NOR gate has two input. both of them are C. produces $S_2 = \bar{C}$

The third NOR gate has two input S_1 and S_2 , produces $S_3 = Y = (A+B)C$
 $S_1 = \overline{A+B}$ $S_2 = \bar{C}$. $Y = \overline{S_2 + S_1} = \overline{\bar{C} + \overline{A+B}}$, according to DeMorgan's law,

$$Y = \overline{\overline{A+B}} \bar{C} = (A+B) \cdot \bar{C}$$

- (2) (7 pts) Draw the complete circuit. Show all connections and label A, B, C, Y, and the intermediate signals.



- (3) (2 pts) Evaluate your circuit for $(A, B, C) = (0, 1, 1)$. Show the output of each NOR gate.

$$\therefore (A, B, C) = (0, 1, 1)$$

$$S_1 = 0$$

$$S_2 = 0$$

$$S_3 = 1 = Y$$

$$\therefore \text{when } (A, B, C) = (0, 1, 1)$$

$$Y = 1$$

Problem 8. (10 pts) **Add two one-bit numbers.**

The inputs A and B are each one bit. Their sum can require two bits: S is the low-order bit and C_{out} is the high-order bit. Thus the result is written as $(C_{\text{out}}S)_2$.

- (1) (5 pts) Complete the table by adding A and B in binary.

A	B	C_{out}	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

- (2) (5 pts) From the table, identify a gate for each output. Draw a circuit that produces both outputs using **one 2-input XOR gate** and **one 2-input AND gate**.

$A, B \rightarrow S$: a XOR gate

$A, B \rightarrow C_{\text{out}}$: a AND gate

The circuit:

